
Reflecting Disk

Consider a large, flat, perfectly reflecting disk of radius $R = 10^8$ m and mass $m = 10^9$ kg. An ideal point light source of power $P = 10^{15}$ W is placed a distance $d = 10^8$ m from the disk's center such that light that hits the disk's center is perpendicular to the disk's surface. If the light source turns on at $t = 0$, find the speed of the disk at $t = 0.4$ s.

The force at radius r on the disk is $\frac{2P}{4\pi(d^2+r^2)c}(2\pi r dr)(\frac{d}{\sqrt{d^2+r^2}})^2 = \frac{rPd^2}{(d^2+r^2)^2c}dr$. (Note that we've projected the component of the light's velocity twice, once to account for the direction of its force and once to account for the number of particles hitting the disk (projected area)). Noting the order of magnitude of the acceleration, we can neglect relativistic effects and the distance that the disk travels. We must note that the light takes some time to travel to the disk, and that different points receive the light at different times. The time it takes the light to reach a radius r is $\frac{\sqrt{d^2+r^2}}{c}$. Noting that this value is greater than t , that means that some parts of the disk have not received light at all. The parts of the disk that have received light are from radius $r = 0$ to $r = \sqrt{c^2t^2 - d^2}$. Thus, the velocity of the disk is

$$\frac{1}{m} \int_0^{\sqrt{c^2t^2 - d^2}} \frac{rPd^2}{(d^2 + r^2)^2c} (t - \frac{\sqrt{d^2 + r^2}}{c}) dr = 1.85 \cdot 10^{-5} \text{ m/s}$$